Lecture 3 - 05/11

Tuesday, 10 November 2026 18:24

Last time: Equivalence of many 3SUM variants…

3SUM Given $S \subseteq [-U, +U]$ are there $a, b, c \in S$ such that $a + b + c = 0$?

3SUM-Finding

Colored-3SUM

3SUM' Given $S \subseteq [0, +U]$ are there $a, b, c \in S$ such that $a + b = c$?

3SUM-with-Duplicates

Small-Integer-3SUM

Target-3SUM

Today: 1. 3SUM hardness for geometric problems (slides)

2. More 3SUM equivalences via additive hashing (on whiteboard)

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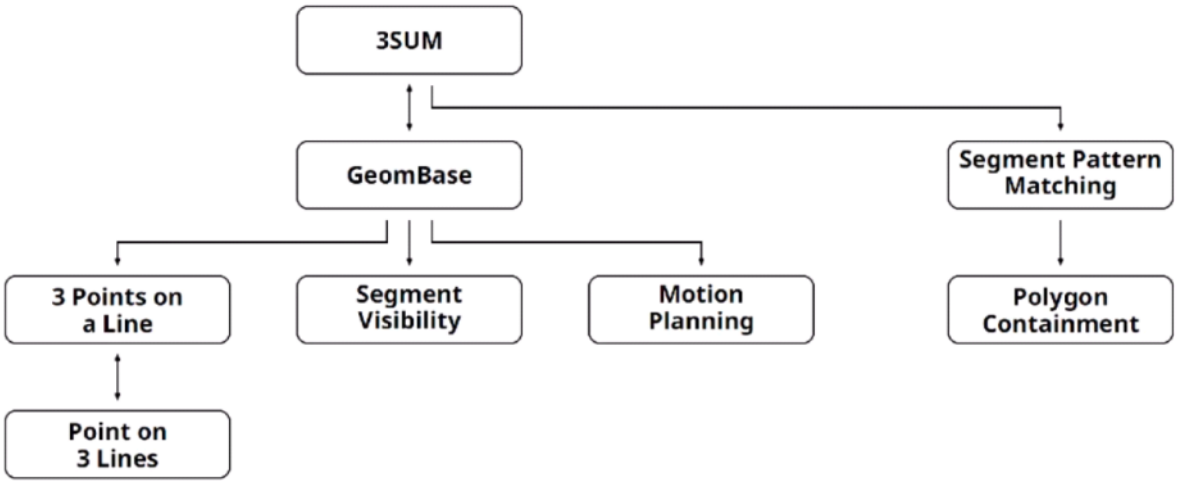
updates:

- simpler 3sum in $O(n^{1.8})$ time
- k-sum for odd k in $n^{\lceil k/2 \rceil - \varepsilon}$

still open: 4-sum in $n^{2-\varepsilon}$

Today: Some 3SUM-Hardness Reductions

[Gajentaan, Overmars '95] [Barequet, Har-Peled '99]

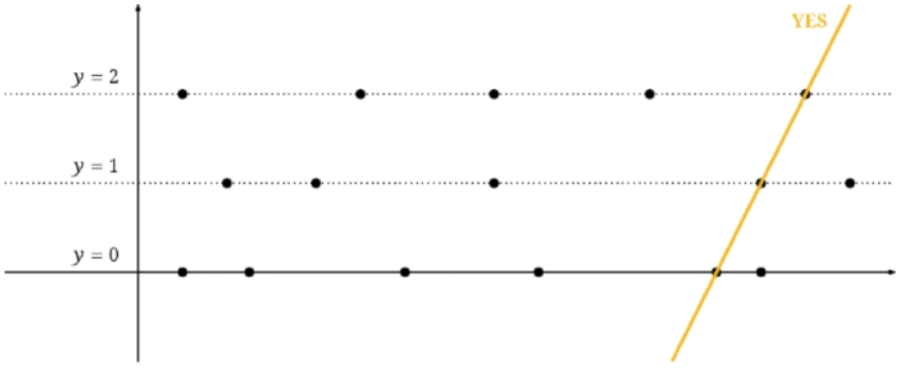


3

3SUM $\longleftrightarrow_2$ GeomBase

GeomBase

Input: n two-dimensional points that lie on the horizontal lines with $y = 0, 1, 2$
Output: $\exists$ non-horizontal line through 3 points?



4

3SUM $\longleftrightarrow_2$ GeomBase

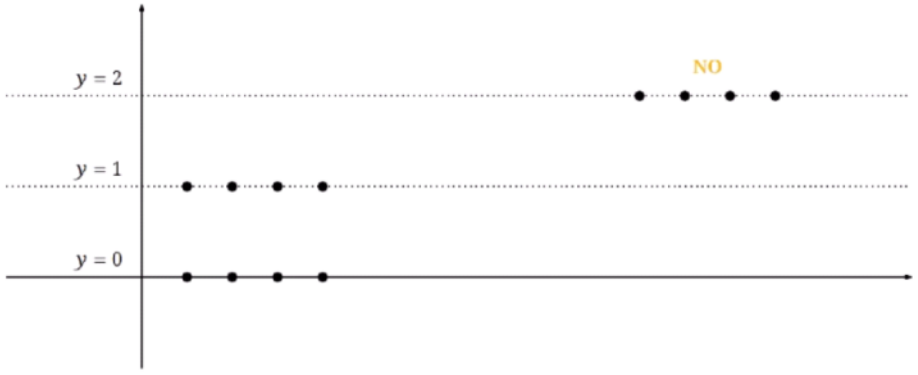
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3SUM $\longleftrightarrow_2$ GeomBase

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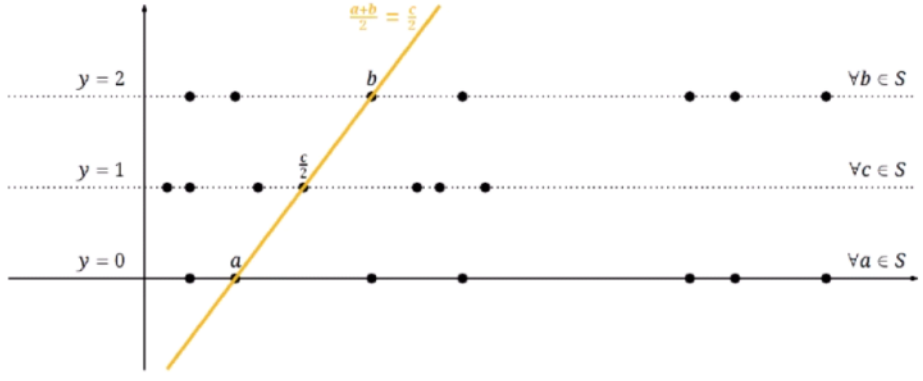


4

3SUM $\longleftrightarrow_2$ GeomBase

GeomBase

Input: n two-dimensional points that lie on the horizontal lines with $y = 0, 1, 2$
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4

3SUM $\longrightarrow_2$ 3 Points on a Line

$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
$x^2 - y^2 = (x - y)(x + y)$

3 Points on a Line

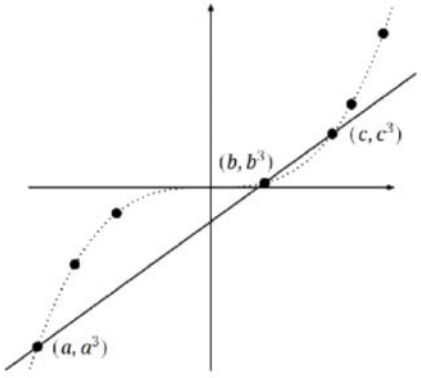
Input: n two-dimensional points
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3 Points on a Line

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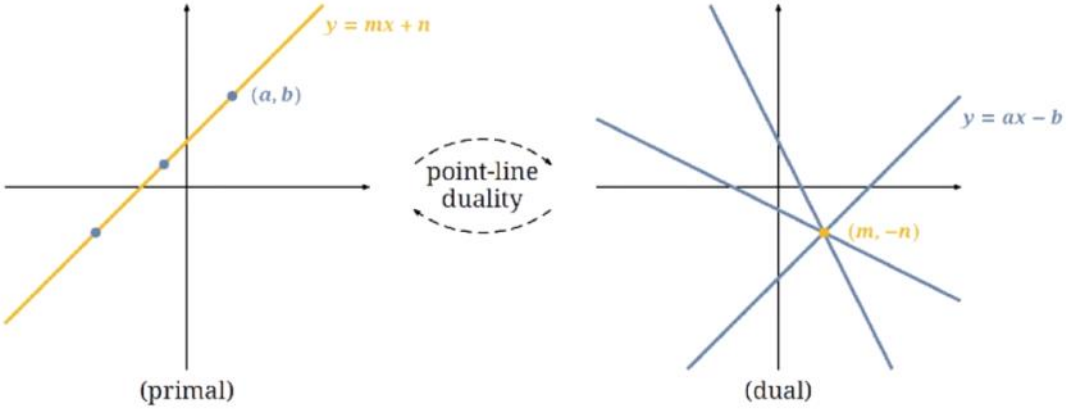


$$\iff \frac{b^3-a^3}{b-a} = \frac{c^3-b^3}{c-b}$$
$$\iff a^2 + ab + b^2 = b^2 + bc + c^2$$
$$\iff a^2 + ab = bc + c^2$$
$$\iff a^2 - c^2 = b(c - a)$$
$$\iff a + c = -b$$
$$\iff a + b + c = 0$$

3 Points on a Line $\longleftrightarrow_2$ Point on 3 Lines

Point on 3 Lines

Input: n two-dimensional lines
Output: $\exists$ point lying on 3 lines?



Visibility Problems

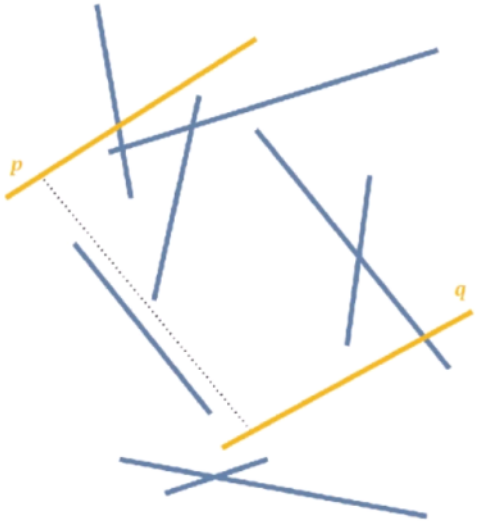
Point-to-Point Visibility
Input: Points p, q, and n opaque line segments
Output: Can p see q?

$O(n)$

Point-to-Segment Visibility
Input: Point p, line segment q, n line segments
Output: Can p see q?

$O(n \log n)$

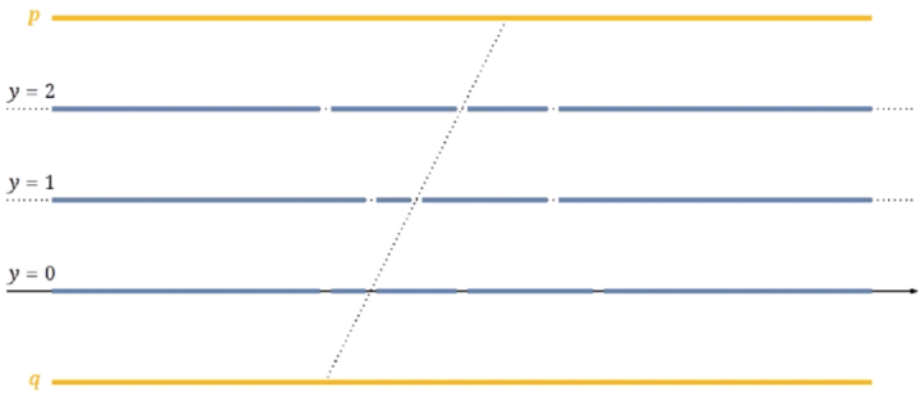
Segment-to-Segment Visibility
Input: Line segments p, q, and n opaque line segments
Output: Can p see q?



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GeomBase $\longrightarrow_2$ Segment-to-Segment Visibility

Segment-to-Segment Visibility
Input: Line segments p, q, and n opaque line segments
Output: Can p see q?



8

Motion Planning Problems

Motion Planning with Translations

Input: Line segments p, q, and n line segments

Output: Can p be translated into q without hitting the obstacles?

"2 degrees of freedom"

$O(n \log n)$

Motion Planning with Translations and Rotations

Input: Line segments p, q, and n line segments

Output: Can p be translated/rotated into q without hitting the obstacles?

"3 degrees of freedom"

3SUM-hard

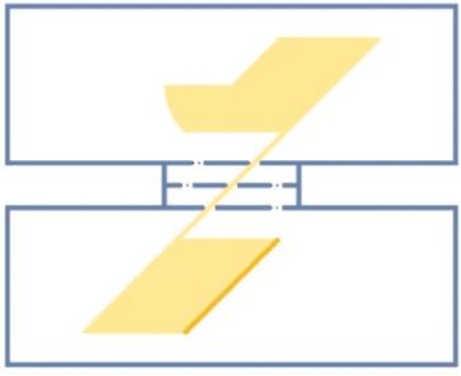


GeomBase $\longrightarrow_2$ Motion Planning

Motion Planning

Input: Line segments p, q, and n line segments

Output: Can p be translated/rotated into q without hitting the obstacles?



Pattern Matching Problems

[Cardoze, Schulman '98] [Barequet, Har-Peled '99]

Point Pattern Matching

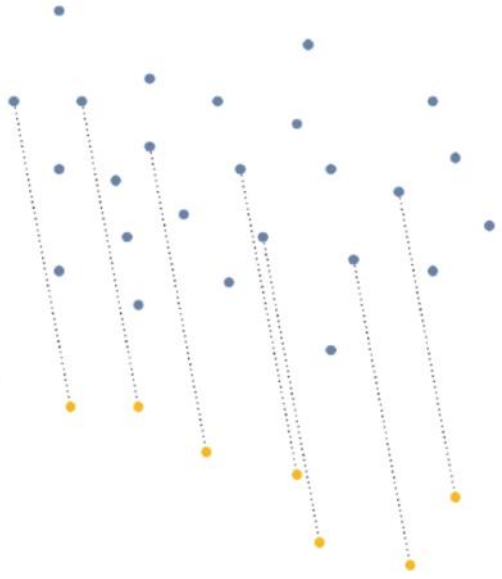
Input: Point sets P and Q of size at most n
Output: Does Q cover a translation of P?

$O(n \log n)$

Segment Pattern Matching

Input: Sets of line segments P and Q
Output: Does Q cover a translation of P?

3SUM-hard



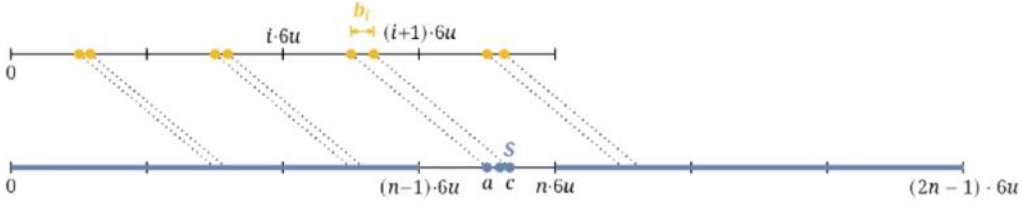
3SUM $\longrightarrow_2$ Segment Pattern Matching

[Barequet, Har-Peled '99]

Segment Pattern Matching

Input: Sets of line segments P and Q of size at most n
Output: Does Q cover a translation of P?

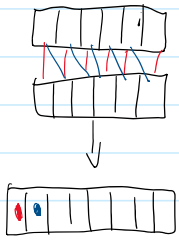
3SUM instance $S = \{b_1, \ldots, b_n\} \subseteq [u]$



small-Integer-3sum: given n integers $S \subseteq [-U, U]$ are there $a, b, c \in S$ s.t. $a+b+c=0$?

• $U = O(n)$, can we find an $O(n \log n)$ alg? FFT

convolution: multiply with shift and sum



input: arrays A, B of len U

output: $C = A \circledast B$ s.t.

$$\forall j \in [2U]: \quad C[j] = \sum_{i=1}^{U} A[i] \cdot B[j-i]$$

equivalent ↓

polynomial multiplication

input: polynomials $p(x) = \sum_{i=0}^{U-1} a_i x^i$

$$q(x) = \sum_{i=0}^{U-1} b_i x^i$$

output: $p \cdot q(x) = \sum_{j}^{2U-1} \left(\sum_{i} a_i b_{j-i} \right) x^j$

naive: $O(U^2)$

FFT: $O(U \log U)$

$O(n + U \log U)$ for Partition-3sum':

- let $A[i] = B[i] = \begin{cases} 1 & i \in S \\ 0 & i \notin S \end{cases}$
- compute $C = A \circledast B$
- "yes" iff $\exists j$ s.t. $j \in S$ and $C[j] \geq 1$

correctness: $C[j] \geq 1 \iff \exists i: A[i] \geq 1 \quad B[j-i] \geq 1 \iff i \in S$ and $j - i \in S$

what about larger U?

$U = O(n^3 \log^3 n)$

Thm: if 3sum with $U = n^3 \lg^3 n$ is in time $T(n)$, then there is an alg that solves 3sum with $U = n^{100}$ in time $O(T(n))$ and succeeds w.p. $\geq 9/10$

$h(x) = (x \bmod p)$

Pf: - pick a random prime p in $[2, n^3 \lg^3 n]$ uniformly

- $S' = \{ h(x) \mid x \in S \}$

  if 3sum$(S') \neq$ "no" return "no"

- return "yes"

→ make 4 calls to 3sum'$(S$, target) with target $= 0, p, 2p, 3p$; return "yes" if one says "yes"

correctness: "no" ⇒ "no" w.p. 1

return "yes" if one says "yes"

correctness: "yes" ⇒ "yes" w.p. 1

$a+b+c=0 \Rightarrow (a+b+c) \bmod p = 0$

"no" ⇒ "no" w.p. $\geq 9/10$

fix a,b,c s.t. $a+b+c \neq 0$

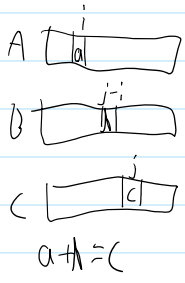
$\Pr[a+b+c=0 \bmod p] = \frac{\#\text{prime divisors of } a+b+c \text{ in } [2, h^3 \lg^3 h]}{\#\text{ of primes in } [2, h^3 \lg^3 h]} \leq \frac{\lg(h^3 \lg^3 h)}{\frac{h^3 \lg^3 h}{\lg h}}$

$= O\left(\frac{\lg h}{h^3 \lg^3 h}\right) = \frac{1}{h^3 \lg h}$

union bound: $\Pr[\exists a,b,c \in S \text{ s.t. } a+b+c=0 \bmod p] \leq h^3 \cdot \frac{1}{h^3 \lg h} < \frac{1}{\lg h} < \frac{1}{10}$

strong 3sum conj: 3sum is hard with $U = h^2$, not solved!

conv-3sum: given arrays A, B, C of length n with numbers in $[1, U]$. are there $i, j \in [n]$ s.t. $A[i] + B[j-i] = C[j]$?



Exercise (will be in HW): 3sum in time $T(n) \Rightarrow$ conv-3sum in time $O(T(n))$

Thm: conv-3sum in time $T(n) \Rightarrow$ 3sum in time $T(n \lg^3 n)$ w.h.p

the idea: conv-3sum basically does 3sum, but it has to be ordered

how to reorder, put a in $A[a], \ldots$

now $i+j-i=j$ and $a+b=c$

but a might be too big, so hash, a in $A[h(a)], \ldots$

want $h(x)$:

- $h(x) = O(n \lg^3 n)$
- $a+b+c=0 \Rightarrow h(a)+h(b)+h(c)=0$
- few collisions

reduction:

- repeat $10^7 \lg n$ times:
  - pick random $p \in [10 n \lg^3 n, 20 n \lg^3 n]$
  - define $h(x) = x \bmod p$
  - set $A[i] =$ random x in S_i

let $S_i = \{x \in S \mid h(x) = i\}$

- define $h(x) = x \bmod p$

- set $A[i] = \begin{cases} \text{random } x \text{ in } S_i \\ +\infty \text{ if } S_i = \emptyset \end{cases}$ let $S_i = \{x \in S \mid h(x) = i\}$

  B, C same way

- if (conv-3sum(A,B,C) = "yes", return "yes"

"no"

correctness:

def: - a is "good" if $|S_{h(a)}| \le 10$

- a is "bad" O.W

- a is "chosen" if $A[h(x)] = a$

same for b, c

claim: if $a+b=c$ and a, b, c "chosen"
then conv-3sum$[A,B,C]$ = yes

pf clear

lemma: $a \neq a' \Rightarrow \Pr[h(a) = h(a')] = \Pr[a - a' = 0 \bmod p] \le \frac{\#\text{prime divisors}}{\#\text{primes}} \le \frac{\log^3}{n \log^2 n} \le \frac{1}{10n}$

$\Pr[a \text{ is "bad"}] \le \frac{1}{10}$

$\Pr[a \text{ is "chosen"} \mid a \text{ is "good"}] \ge \frac{1}{10}$

$\Pr[a \text{ is "chosen"}] \ge \frac{9}{100}$

$\Pr[a, b, c \text{ "chosen"}] \ge \left(\frac{9}{100}\right)^3$

while not that small, its constant
we repeat 10^7 times for that